

Extensions of HyperPose: Echo, Savant, Loop

Echo, Savant, Loop

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ABSTRACT

This presentation introduces several solutions which came as byproduct from our work on HyperPose. Echo is a technique for applying stylistic modifications to select poses in character animation, and extending these effects to unaltered data. Savant is a novel approach to generating runtime secondary motion for character cloth, hair, muscles, props, etc. Loop describes an improved method for mocap aimed to serve next generation of realism in character animation

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1 ECHO

This talk comes as an addendum to HyperPose talk (Berezhnyak, 2024), so we assume the concept of Principal Dynamic Poses to be already known to the audience. The key aspect of this method is its ability to capture a representative set of features from the overall motion data. This implies that any modifications or additions to these features can be extrapolated to the entire dataset, as illustrated in Figure 1.

Multiple methods have been proposed in the past for segmentation of motion sequences, or identification of relevant poses in the data (Matsuda and Kondo, 2004) (Miura et al., 2014) (Tanuwijaya and Ohno, 2010). It appears only natural to compare such poses; and if such sparse samples can represent the whole, then modifications introduced to them will represent modified whole. A significant segment of this talk is devoted to presenting the “Body Space Transform (BST)” method. This approach is pivotal for accurately determining the positioning, rotation, and velocities of body parts in the context of the pose as a whole.

We cover our method of eliminating redundancy in motion, which allows representing multiple long sequences as one short clip the animators will be able to reason with: motion distillation. This allows tracking the style spanning multiple minutes or hours of data from one short, animator-friendly timeline and is essential for preserving artistic agency. This builds directly on the concept

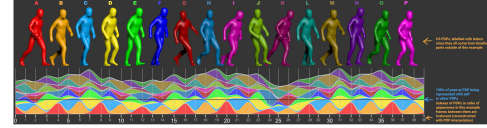


Figure 1: PDPs and their effects over a timeline sample. Notice the cadence of A-C-B-D; in the data itself, this represents the walk cycle (including turns, leans, etc.) It is worth noting that any frame on this timeline can be described fully using no more than 3 PDPs from the set above.

of arranging families of animations based on their parametrization (Buttner, 2015).

The distilled sequence can then be used to add stylistic changes to key PDPs, or simulate muscles, cloth, etc. Once such extra data per PDP is generated, it can be propagated to the whole set of motions, blending weights of “solved” PDPs over all untouched poses.

We also cover the UI side of Echo as well as the necessary pipeline modifications for the integration. We conclude with an in-depth dive into the known functional limitations and their possible solutions, and future steps for both user-facing and performance optimization side of Echo.

2 SAVANT

In the realm of video game character animation, the foundational approach of using polygons, vertices, and a hierarchy of transforms (anthropomorphized as “bones” or “joints”) was standardized in the 1990s. This method, supporting various techniques such as keyframing, motion capture, and physics-based systems, results in joint-driven vertex transformations that, interpolated over time, emulate motion. Although not the most mathematically efficient, this method’s widespread utility can be attributed to its alignment with human cognitive processes.

Secondary animation, however, has seen a multiplicity of approaches, each finding niche applications dependent on factors like game genre, platform capabilities, level of detail requirements, and more. Yet, we argue that existing solutions fall short of delivering a holistic balance of fidelity, control, and performance.

Savant introduces an innovative methodology in the generation of high-fidelity, cost-efficient secondary motion for characters in real-time, targeting traditionally challenging aspects such as dynamic cloth, hair/fur, props, and nuanced muscle/skin deformations.

This presentation begins by outlining Savant’s high-level concept and reasoning approach, followed by a detailed analysis of the existing problem space and the strengths and weaknesses of

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current methods. We delve into the motion-informed generation of topology and skin data and the methods of parsing extensive animation datasets to identify blend points for describing any desired pose.

Further, we showcase Savant’s capabilities in handling large- and small-scale details, layered cloth dynamics, physics interactions, character constructors and cloth combinations, cross-fitting of clothing across varying character proportions, and the scalability of its performance and memory footprint.

The talk concludes with a discussion on areas of concern and a roadmap for future developments of Savant.

3 LOOP

Loop is a mathematical method for creating actor movement patterns for various use cases (locomotion, prop interaction, etc.) We solve for the best possible path for the actor to follow during the game mocap session to capture the most versatility with the least redundancy.

Higher motion versatility is useful to avoid “motion tiling” and provide more possible means to achieve goals, improving realism. Lower redundancy allows for shrinking of the data footprint. Optimization of capture time allows more meaningful blocks of actions with a precise focus on current goals and more effective planning of schedules.

In this talk, we discuss the current industry standard mocap scheme approach, explain the reasoning for improvement, and elaborate on the logic and math of utility. We start with a description of the problem space, draw parallels with human trajectories captured in real-life scenarios, identify key patterns to replicate, and

present an algorithm for the capture of the most economical motion patterns.

We incorporate motion types (walking, running, etc.), motion modifiers (which foot forward, facing direction, speed), interactions (world, prop, self, other character), and discuss known constraints of motion capture sessions. Our resulting solution collapses the shot list into a formula that is easy to scale both for more types of actions and for higher versatility of the same actions.

We provide a DIY recipe for easy replication or adaptation for specific needs. We also elaborate on the stage setup and direction for the “capture staff (Guides)” and “capture talent (Actor).”

Finally, we offer a preliminary comparison of results achieved using the classical method and Loop. The limited data we have suggests that Loop provides double the motion versatility in half the capture time. Further research and a larger sample set are necessary to evaluate the scope of improvement with more precision.

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